

Three Macs and the Speed of Light

Working out, from the ground up, why two of your machines can share a brain and the third one never can.

Measured on Mac mini M4, 16 GB macOS 26.5.2 4 August 2026

01 **A model is a pile of numbers.**

Qwen3-32B is thirty-two billion numbers. Each one is squeezed into four bits, which is half a byte, so the whole model is about 16 GB. That is all it is. Not code, not a program that thinks. A very large file full of numbers that were settled during training and never change again.

02 **Every word it writes reads all of them.**

To produce one token, roughly one word, the model takes your text and multiplies it through every one of those numbers, in order, once each. Then to produce the *next* word, it does the entire thing over again. All thirty-two billion, every time.

THEREFORE

Writing a sentence means hauling 16 GB out of memory and into the chip, once per word.

03 **So speed is a plumbing problem, not a maths problem.**

Your GPU can do trillions of calculations per second. It can only *read* about 120 GB per second. So the arithmetic finishes almost instantly and then sits there waiting for the next batch of numbers to arrive. The chip is not thinking hard. It is queuing.

Which gives a prediction you can check: speed should be roughly memory bandwidth divided by model size. I ran a small model on your mini. It occupies 1.035 GB, so $120 \div 1.035 = 116$ words per second in the ideal case. It actually managed 72.7. The gap is normal, since nothing ever hits its rated bandwidth, but the shape of the answer holds.

THEREFORE

A more compressed model is not merely lighter. It is proportionally faster, because there are fewer bytes to drag.

04 **But the wall you actually hit is whether it fits at all.**

Your mini has 16 GB of memory, and macOS only lets the graphics side claim about 10.6 GB of it. A model that needs sixteen does not load slowly. It does not load.

This is the real reason anyone connects Macs together. Not to go faster. To have somewhere to put the thing.

05 **There are two ways to cut a model in half.**

Picture the model as sixty-four pancakes in a stack. Each pancake is one layer, and they are all the same shape.

You can cut *horizontally*: machine A keeps the bottom thirty-two pancakes, machine B keeps the top thirty-two. Work climbs up through A, gets handed over, climbs up through B.

Or you can cut *vertically*: both machines keep all sixty-four pancakes, but each holds only the left or right half of every one. They work side by side, layer by layer.

These sound like the same idea wearing different hats. The rest of this note is about how violently they differ.

06 **Cutting vertically makes the machines wait for each other.**

If each machine holds half of a layer, then when each finishes its half, each is holding half of an answer. Neither one can take a step forward. They have to stop, send their halves across, and add them together.

That happens twice inside every layer. Sixty-four layers, so 128 full stops for every single word.

Cut horizontally instead and machine A simply finishes its thirty-two pancakes, hands B a small parcel, and B carries on. One stop per word.

THEREFORE

The vertical cut costs 128 waits per word. The horizontal cut costs one.

07 **And waiting has nothing to**

do with how fat your pipe is.

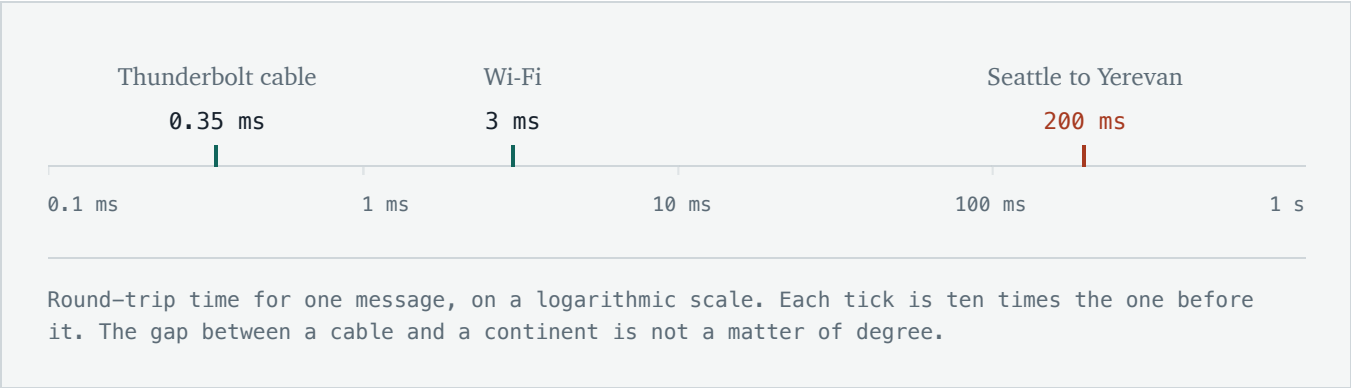
The parcel being sent is tiny, around 8 KB. Over a Thunderbolt cable, physically pushing those bytes onto the wire takes under 2 microseconds. The round trip takes about 300.

So ninety-nine percent of the wait is not moving data. It is the message travelling there, and the reply travelling back. Doubling your bandwidth halves the two microseconds and leaves the three hundred exactly where they were.

THEREFORE

A wider pipe buys you nothing here. Only a shorter one does. What you are paying for is distance.

08 So let us measure the distances.



Now multiply each of those by 128, because that is how many times per word the two machines have to stop and check in with each other.

LINK	ROUND TRIP	WAITING, PER WORD	CEILING
Thunderbolt cable	0.35 ms	45 ms	22 words/s
Wi-Fi	3 ms	384 ms	2.6 words/s
Seattle to Yerevan	200 ms	25.6 s	0.04 words/s

The last row is twenty-five seconds of waiting to produce one word. Not one answer. One word. Then it starts waiting again.

09 **That last number is physics, not a configuration mistake.**

Seattle to Yerevan is about 10,300 km. Light inside glass fibre travels at roughly two-thirds of its speed in a vacuum. Even with a perfect, straight, dedicated cable that nobody has ever built, the round trip cannot go below 103 ms. Real traffic detours through Frankfurt and a dozen routers, so it lands near two hundred.

There is no faster tier you can buy. You would have to move Armenia.

10 **And I measured the cost of waiting with the distance set to zero.**

I ran the same small model on your mini twice. Once normally. Once split in half across two processes on *the same machine*, so there is no cable, no room, no ocean. The network is a loop inside the operating system, which is as fast as a network can possibly be.

It went from 60.2 words per second to 28.6. Splitting it made it 2.1× slower, with the distance removed entirely.

THEREFORE

Never split a model that already fits. Splitting is a cost you accept, never a speed-up you chase.

11 **Which settles the arrangement.**

The mini and the laptop, joined by a Thunderbolt cable, share one model.

Splitting doubles the memory you can read from, and the waiting eats most of that gain back, so you end up roughly where a single machine would have been if it could have held the model. It could not. That is the whole trade: you are not buying speed, you are buying room.

The M3 Max in Armenia runs its own model, whole, by itself. You send it a question and pay the two hundred milliseconds once, instead of a hundred and twenty-eight times per word. For anything that is not live conversation, that delay is invisible. It is very likely the most capable machine of the three, and it has plenty of useful work to do. It simply cannot be part of the same brain.

A cluster is not a machine for going fast. It is a machine for making something possible that was not, and the only thing it asks in return is that its members stay close together.